

Freertos中FAT文件系统的制作说明文档

1. freertos文件系统介绍

Ramdisk介绍:

Ramdisk实际上是从内存中划出一部分作为一个分区使用（用内存模拟flash），使用时我们创建一个ramdisk作为根目录"/"的存储空间，在根目录下创建挂载点。

FAT文件系统挂载介绍:

使用文件系统前，先确定分区中是否存在FAT16/32或exFAT文件系统，如果存在将分区挂载出来，不存在就格式化擦除原来的内容，并在存储介质上新建文件分区表和目录，用来记录数据存放的偏移地址和剩余空间。数据以文件的形式存储，写入新文件时，先在目录中创建一个索引，指示文件存放的偏移地址。当读取数据时，从目录中找到该文件的索引，进而在相应的地址中读取数据。

注：目前支持 nor和mmc挂载fat文件系统。

2. 创建和修改文件系统

1. 创建空的文件系统（以4M为例）

```
mkfs.fat -S 4096 -C fat-test.img 4096
```

参数解释:

- S logical-sector-size 逻辑扇区的大小
- C 创建目标文件，用此选项必须给出<block-count>单位是kBytes

命令效果：在当前目录产生一个img文件，这是一个空镜像

2. 修改文件系统内容

```
mkdir fs_tmp      #创建文件夹
sudo mount -t vfat fat-test.img    #挂载文件系统
. . . . .
. . . . .
. . . . .          #修改文件系统，添加或者删除文件
sudo umount ./fs_tmp    #取消挂载
```

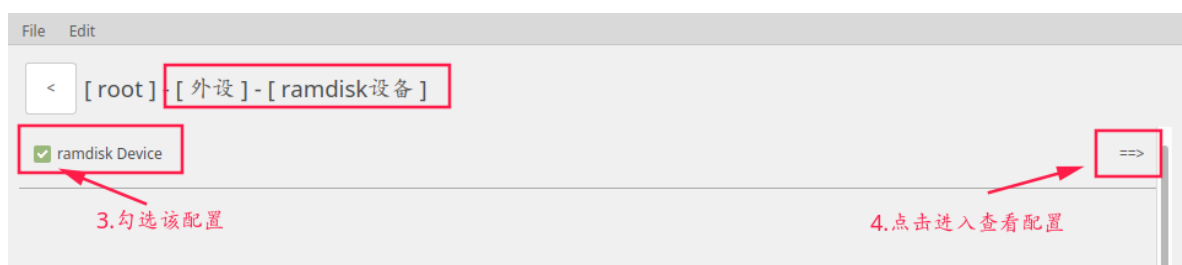
3. nor配置流程

3.1 配置ramdisk设备

1. 从主界面进入"外设"配置，找到ramdisk设备配置



2. 勾选ramdisk设备，查看设备配置，保持默认即可





3.2 配置设备文件系统

1. 从主界面进入设备文件系统

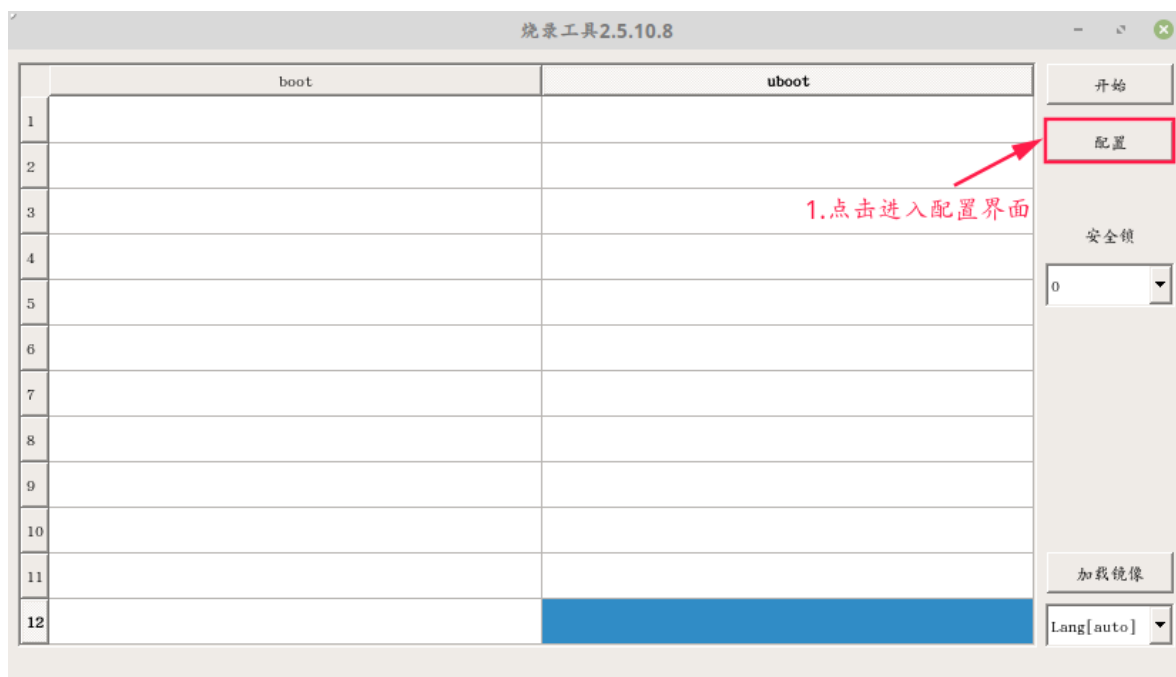


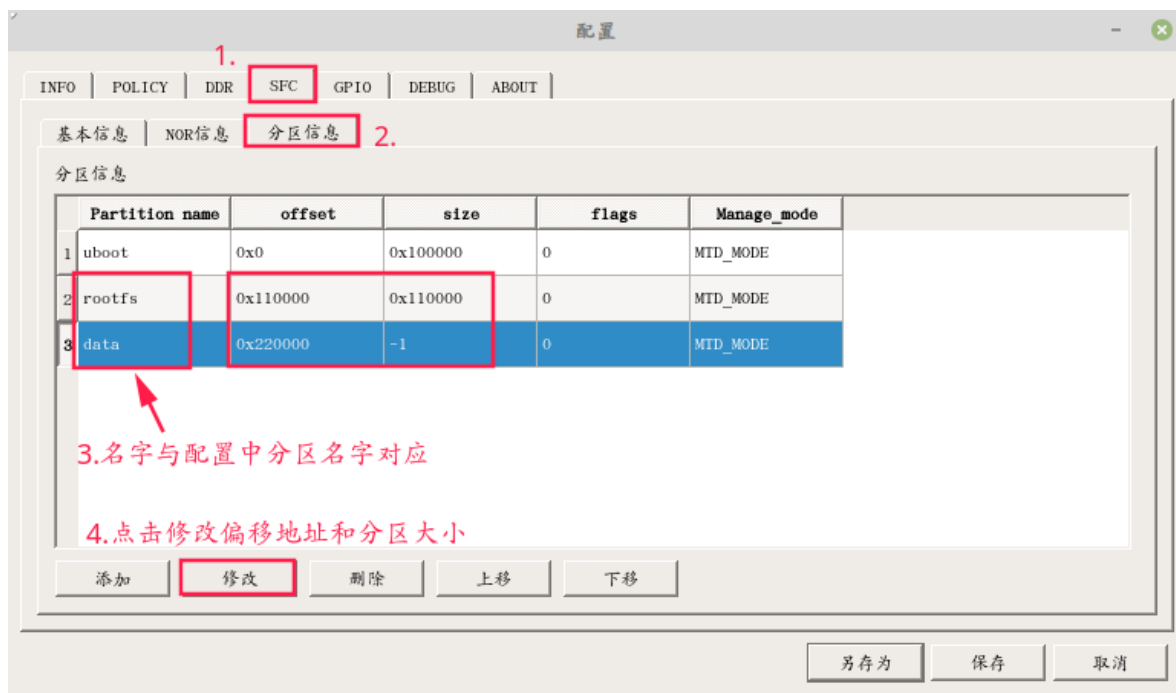
2. 添加fat文件系统挂载分区





3.3 配置烧录工具，设置分区偏移地址与分区大小





烧录时会把分区信息烧录到存储设备。启动会读取分区信息，并与FAT文件系统挂载名称匹配，匹配成功就将对应分区挂载到根目录下。

挂载情况：

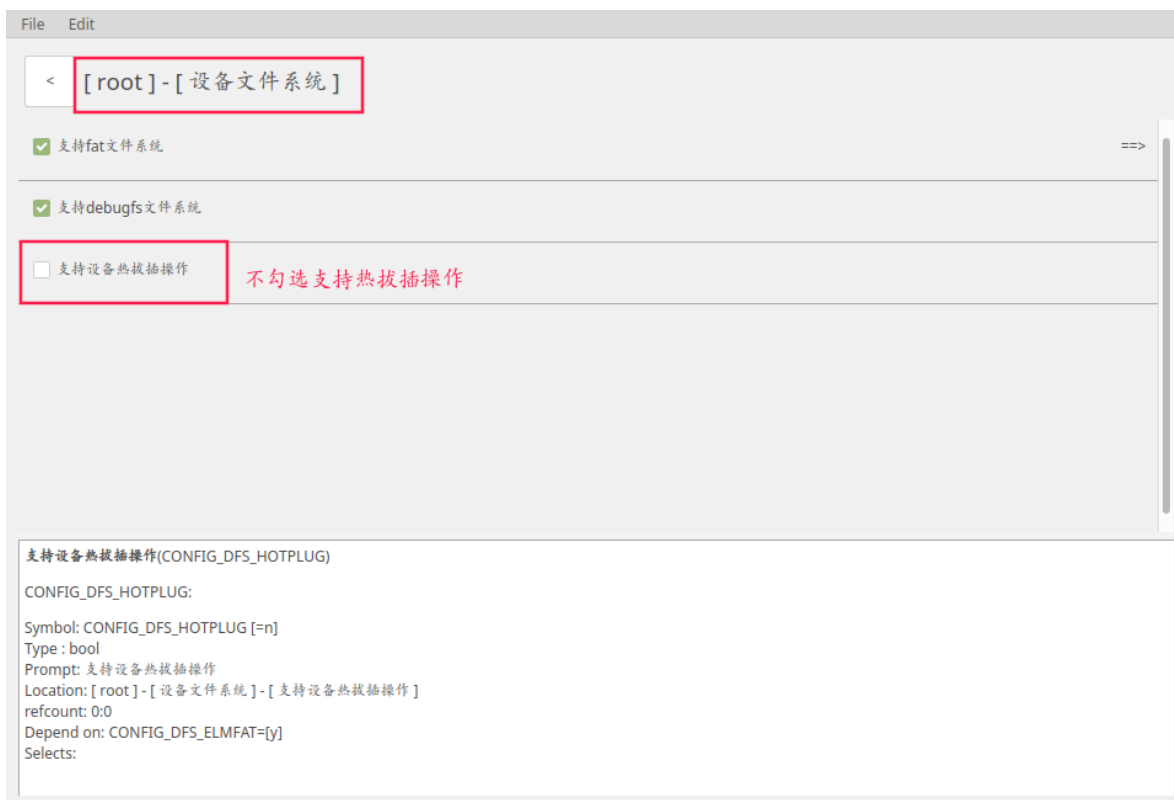
```
$ ls
Type/Size      Name
<DIR>          rootfs
<DIR>          data
$
```

4. mmc配置流程

4.1 不可拔插设备配置

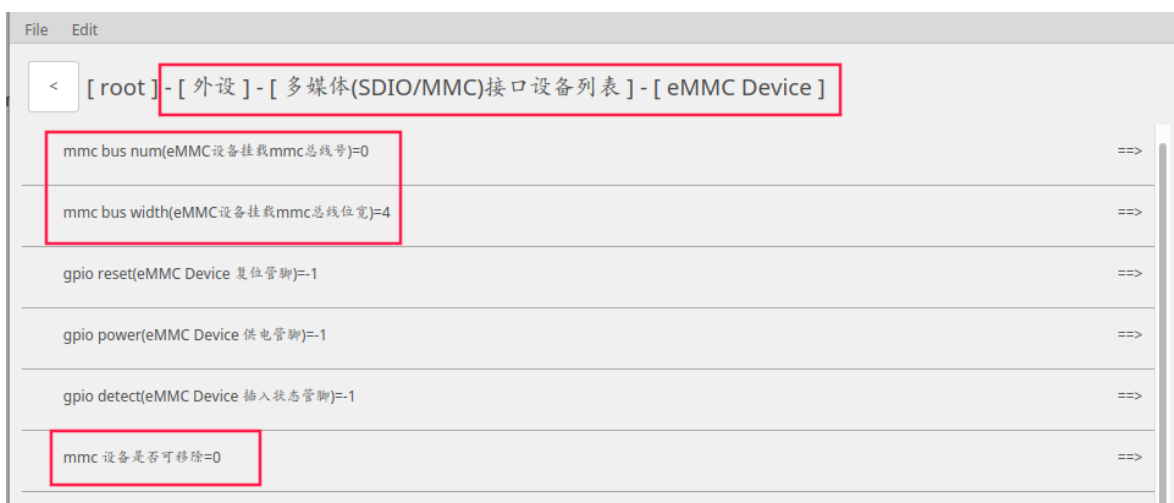
4.1.1 emmc flash

1. 配置ramdisk设备，同3.1配置流程
2. 配置不勾选支持热拔插操作，MMC配置中勾选设备不可移除。



3. 修改MMC总线和接口配置

(注：MMC总线要与eMMC Device中设备挂载的总线号对应)



4. 启动后会使用默认的分表，如需更改分表信息，可修改如下代码中的参数。

freertos/devices/mmc/mmc_partition.c

static struct block_device_partition *get_emmc_device_default_partition_info(void)

```

/* partition-0 */
uint64_t start_offset = 99 * 1024 * 1024; /* 99MByte */
uint64_t part0_size = 2 * 1024 * 1024; /* 2MByte */

sector_size = emmc_device_get_partition_sector_size(start_offset, card->card_blk_size);
index = 0;
parts[index].name      = alloc_partition_name("rootfs");
parts[index].offset     = start_offset; /* 99M */
parts[index].mask_flags = PART_FLAG_RDWR | PART_TYPE_BLK;
parts[index].size       = part0_size; /* 2M */
parts[index].sector_size= sector_size;

/* partition-1 */
start_offset = start_offset + part0_size;
uint64_t part1_size = (card->card_capacity << 10) - 1; /* Unit: Byte */
//part1_size = part1_size - start_offset;
part1_size = 8 * 1024 * 1024; /* 8M */

sector_size = emmc_device_get_partition_sector_size(start_offset, card->card_blk_size);
index = 1;
parts[index].name      = alloc_partition_name("data");
parts[index].offset     = start_offset; /* 101M */
parts[index].mask_flags = PART_FLAG_RDWR | PART_TYPE_BLK;
parts[index].size       = part1_size; /* 8M */
parts[index].sector_size= sector_size;

```

4.2 可动态拔插设备配置

(根据MBR/GPT分区表自动挂载设备/分区)

4.2.1 TF卡

1. 配置ramdisk设备, 同3.1配置流程
2. 从主界面进入设备文件系统配置界面,勾选支持热拔插操作



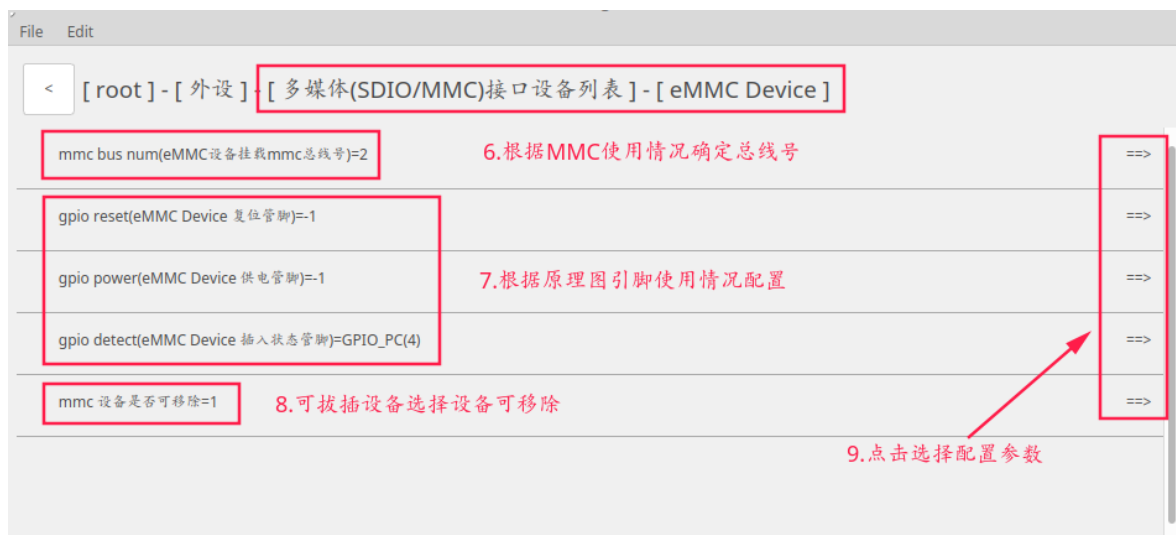
3. 配置MMC接口设备

从主界面进入外设选项

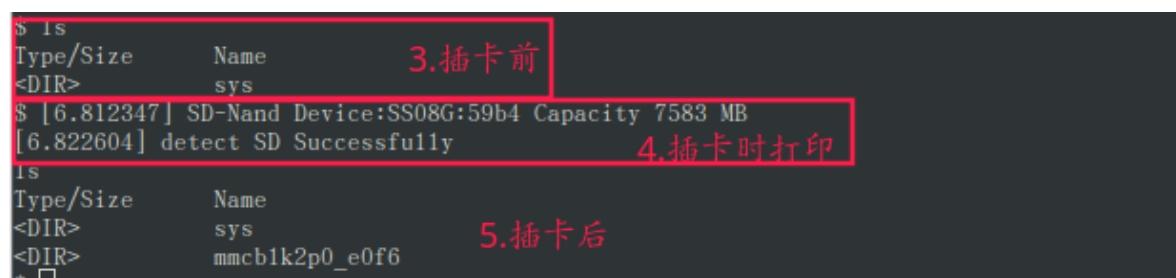


找到MMC接口配置





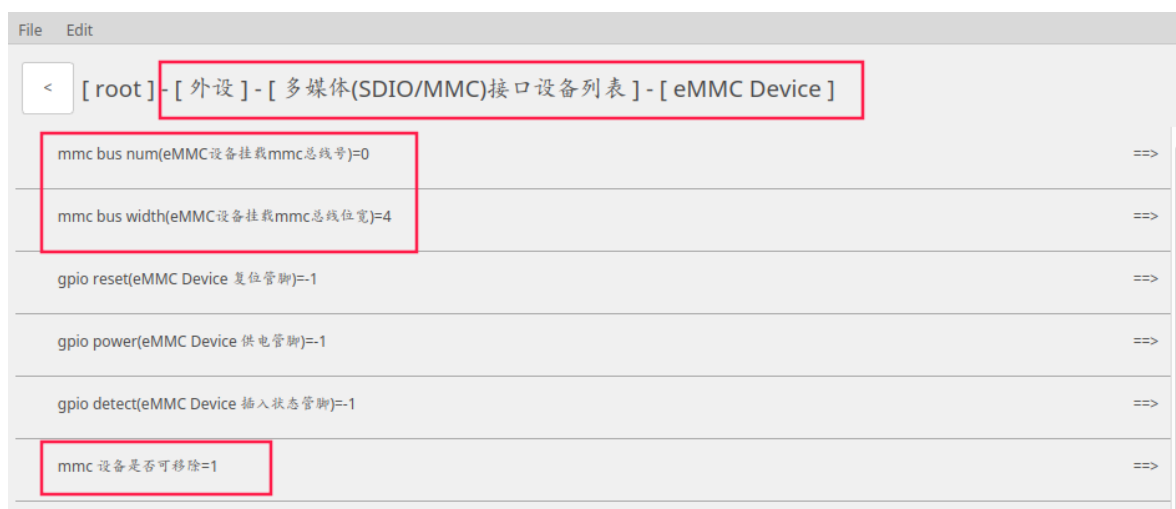
3. 编译烧录之后(如图为TF卡探测效果)



4.2.2 emmc flash(不推荐按照可拔插设备配置)

1. 其他配置同TF卡配置，修改MMC接口配置

(注：MMC总线要与eMMC Device中设备挂载的总线号对应)



2. 打开以下宏定义，可以打印出GPT分区信息

```

ces_emmc.c  C mmc_partition_parse_mbr_gpt.c X  C dfs.c
devices > mmc > C mmc_partition_parse_mbr_gpt.c > ...
You, a few seconds ago | 4 authors (YangHuanHuan and others)
1  #include <malloc.h>
2  #include "mmc_devices.h"
3  #include "mmc_partition.h"
4  #include "mmc_partition_gpt.h"
5  #include "mmc_partition_mbr.h"
6  #include "mmc_partition_fat.h"
7  #include <ctype.h>
8  #define DEBUG_MBR_GPT_PARTITIONS
9  #define DEBUG_PARTITION_PARAMS

```

```

[0.054138] GPT Header:
[0.054216] Signature           : EFI PART
[0.054386] Version            : 0x00010000
[0.054563] Hdr Size              : 92
[0.054713] Hdr CRC32          : 0x9663c367
[0.054890] Reserved              : 0x00000000
[0.055066] Hdr Start LBA      : 1
[0.055213] Backup Hdr Start LBA : 7634943
[0.055380] Partition Start LBA    : 34
[0.055530] Partition End LBA      : 7634943
[0.055696] Partition Table Start LBA : 2
[0.055843] Number of Partition Entry : 7
[0.055990] Size of Partition Entry   : 128
[0.056143] Partition Table CRC32  : 0xbbf04dfb
[0.056320] GUID                    : B8EA79C0-4E14-EB1D-E052-8E1979D11D64
[0.056583] GPT Sector Size(Default) : 512
[0.056736] Reserved2 must be all 0
[0.056860]
[0.057050] Partition Table:
[0.057145] [Num]  Partition Type GUID          Unique Partition Guid          Start(sector)    End(sector)    Size    Name
[0.057612] [ 0]  EBD0A0A2-B9E5-4433-87C0-68B6B72699C7  4CBF775E-88B8-842A-720E-8F1AA7423E22      2048          18431    8388608    kernel
[0.058239] [ 1]  EBD0A0A2-B9E5-4433-87C0-68B6B72699C7  1254DC54-DBFA-A97A-3AD9A0EE956D      18432          223231   104857600   rootfs
[0.058857] [ 2]  EBD0A0A2-B9E5-4433-87C0-68B6B72699C7  B67346DB-4390-1162-5F9F-15D5D6B3CC56      223232          239615    8388608    kernel2
[0.059477] [ 3]  EBD0A0A2-B9E5-4433-87C0-68B6B72699C7  1AFF183D-5755-0EF8-E734-0B9F283D2984      239616          444415   104857600   rootfs2
[0.060097] [ 4]  EBD0A0A2-B9E5-4433-87C0-68B6B72699C7  CF2CB10B-2FD4-E1E7-4FD0-308E45AED533      444416          446463    1048576    ota
[0.060705] [ 5]  EBD0A0A2-B9E5-4433-87C0-68B6B72699C7  26662935-3656-16FF-6AEB-50D1721A01BE      446464          7634943   3680501760   userdata
[0.061274] Mount Partition Param
[0.061386] Name      offset      size      sector_size    mask
[0.061610] mmcblk0p0  0x100000    0x800000    0x000000200    0x000000013
[0.061909] mmcblk0p1  0x900000    0x6400000   0x000000200    0x000000013
[0.062209] mmcblk0p2  0x6d00000   0x800000    0x000000200    0x000000013
[0.062509] mmcblk0p3  0x7500000   0x6400000   0x000000200    0x000000013
[0.062809] mmcblk0p4  0xd900000   0x100000    0x000000200    0x000000013
[0.063109] mmcblk0p5  0xda00000   0xdb600000  0x000000200    0x000000013
[0.063636] vendor init...

```

3. 从GPT分区表读到的分区如果不是FAT格式，会挂载失败，出现以下打印是正常打印(不推荐将emmc flash配置成可拔插设备的原因)

```

[0.063636] vendor init...
[0.064258] filesystem:mount Device(mmcblk0p0) Type(elm) to DIR(/mmcblk0p0) failed. ret -1 errno -2
[0.064592] mount file system [mmcblk0p0] failed. partition type not FAT/FAT32/exFAT unsupport
[0.065076] filesystem:mount Device(mmcblk0p1) Type(elm) to DIR(/mmcblk0p1) failed. ret -1 errno -2
[0.065410] mount file system [mmcblk0p1] failed. partition type not FAT/FAT32/exFAT unsupport
[0.065878] filesystem:mount Device(mmcblk0p2) Type(elm) to DIR(/mmcblk0p2) failed. ret -1 errno -2
[0.066210] mount file system [mmcblk0p2] failed. partition type not FAT/FAT32/exFAT unsupport
[0.066675] filesystem:mount Device(mmcblk0p3) Type(elm) to DIR(/mmcblk0p3) failed. ret -1 errno -2
[0.067007] mount file system [mmcblk0p3] failed. partition type not FAT/FAT32/exFAT unsupport
[0.067471] filesystem:mount Device(mmcblk0p4) Type(elm) to DIR(/mmcblk0p4) failed. ret -1 errno -2
[0.067803] mount file system [mmcblk0p4] failed. partition type not FAT/FAT32/exFAT unsupport
[0.068264] filesystem:mount Device(mmcblk0p5) Type(elm) to DIR(/mmcblk0p5) failed. ret -1 errno -2
[0.068596] mount file system [mmcblk0p5] failed. partition type not FAT/FAT32/exFAT unsupport

```

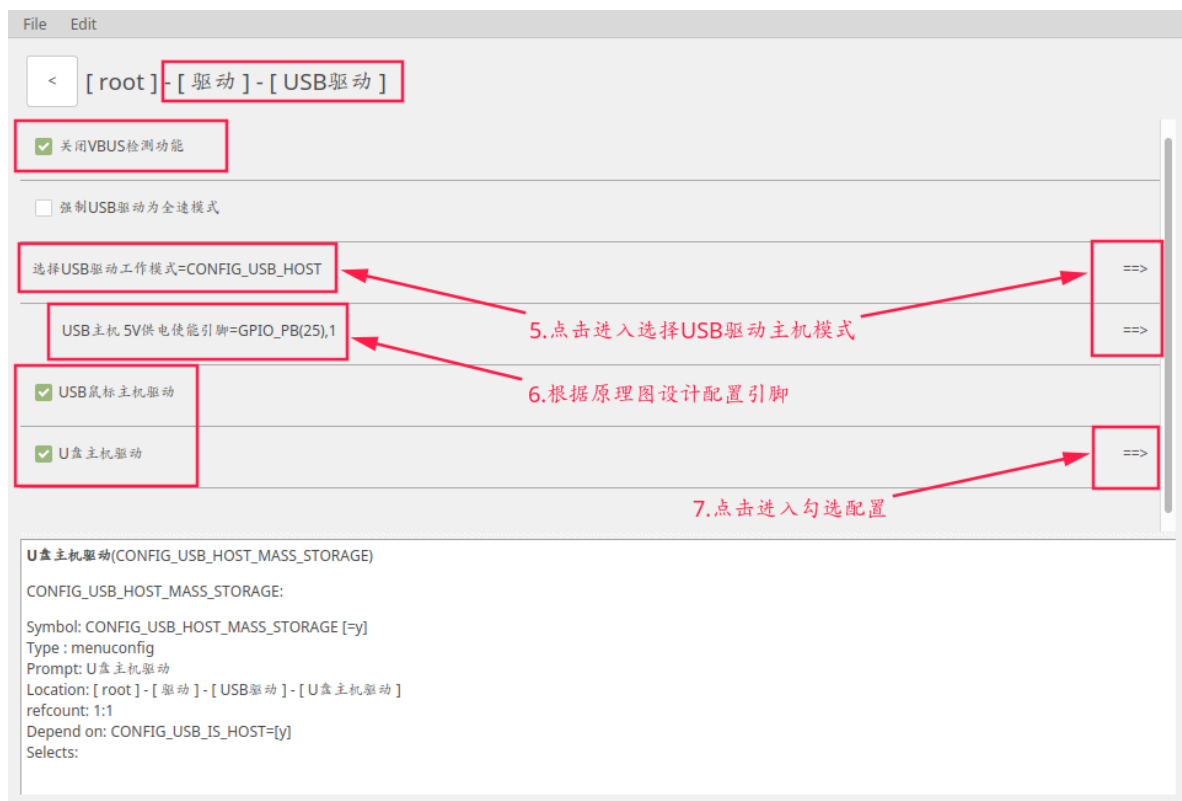
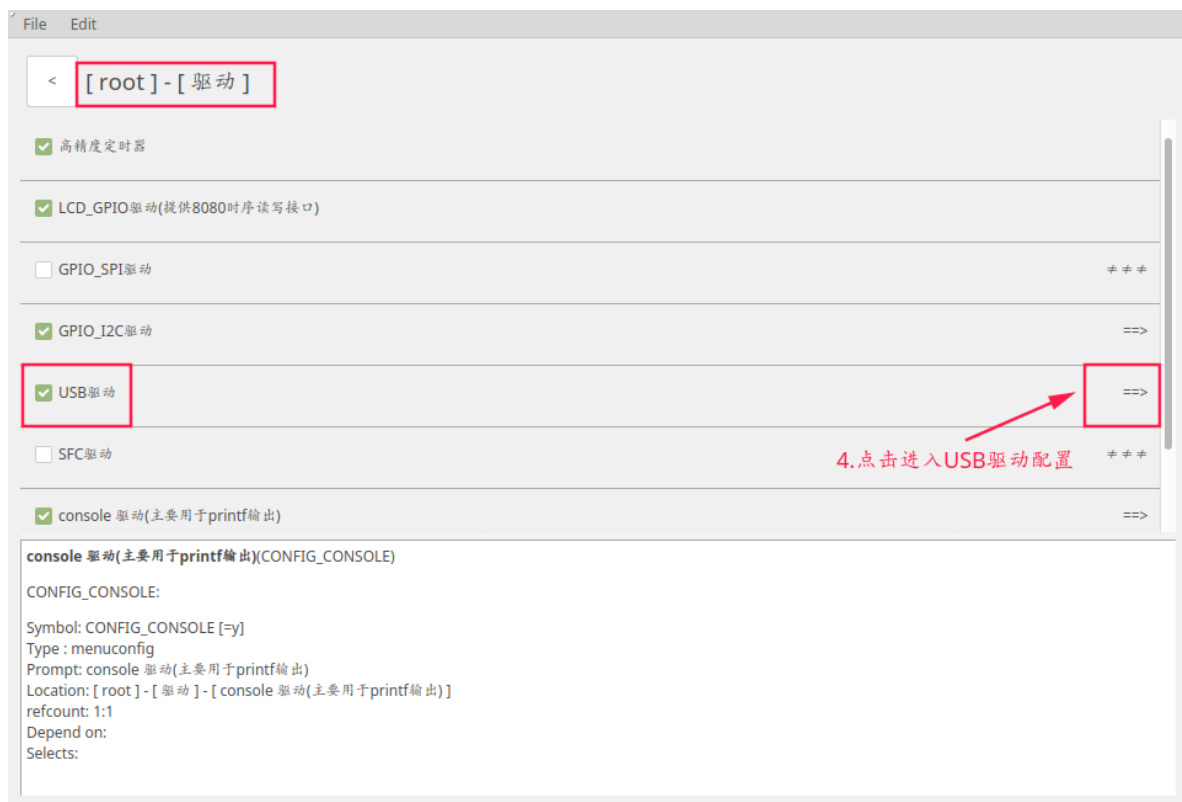
5. U盘配置流程

1. 从主界面进入设备文件系统配置界面,勾选支持热拔插操作



2. 回到主界面进入驱动配置





3. 编译烧录之后(如图为U盘探测效果)

```
$ [186.140819] ++OTG Interrupt gotgint=80000 [a_host]
186.145329] ++OTG Interrupt: Debounce Done++
186.256496] new high speed USB device using and address 2
186.383163] New USB device devnum 2, busnum 1
186.387240] New USB device found, idVendor=ffff, idProduct=5678
186.392960] New USB device strings: Mfr=1, Product=2, SerialNumber=3
186.399114] Product: Disk 2.0
186.401887] Manufacturer: USB
186.404661] SerialNumber: 4526971256615384643
186.408822] configuration #1 chosen from 1 choice
186.413360] devnum 2: adding (config #1, interface 0)
186.418202] Transport: Bulk
186.420954] Protocol: Transparent SCSI
186.424744] USB Mass Storage Max LUN 0
186.428744] Vendor: VendorCo
186.431522] Product: ProductCode
186.435161] Version: 2.00
186.437761] Type: Direct-Access ANSI SCSI revision: 04
186.443569] Removable: 1
186.446488] Block Size 512, Block Count 122880000
186.452976] MBR partition
```

插入后打印U盘信息

```
$
$ ls
Type/Size      Name
<DIR>          rootfs
<DIR>          data
<DIR>          svs
<DIR>          udisk0p0
```